

University of Wollongong

Enabling Mathematics For Engineers

DPEN021 Enabling Mathematics For Engineers – Trigonometry & Calculus Mock Exam

Pages	Total Marks	Section	Questions
14	50	3	24
Section 1 Multiple Choice		10 questions	10 total marks
Section 2 Short Answer		10 questions	20 total marks
Section 3 Long Answer		4 questions	20 total marks

Name _____

Student ID _____

Date _____

Signature _____

How to use this exam

- 1 Set a timer. Practice under real exam conditions! No notes, no AI assistance.
- 2 When finished, scan the QR code on the last page to submit and see the correct answers with detailed explanations.
- 3 Review your mistakes carefully and note down areas that need improvement. Use Studocu AI to create targeted study materials like quizzes, summaries, and study guides.

Multiple Choice

Each question is worth 1 mark. Select the most appropriate response.

Question 1

1 mark

A circle has a radius of 2 cm. What is the area of a sector in this circle that subtends an angle of 1.2 radians at the center?

☐ A 4.8 cm^2 ☐ B 1.2 cm^2 ☐ C 2.4 cm^2 ☐ D 0.6 cm^2

Question 2

1 mark

What is the period of the trigonometric function $y = 3 \sin(2x)$?

☐ A "2"☐ B π ☐ C 2π ☐ D $\frac{\pi}{2}$

Question 3

1 mark

Determine the derivative of $f(x) = \cos(3x)$ with respect to x .

☐ A $3 \sin(3x)$ ☐ B $-3 \sin(3x)$ ☐ C $-\sin(3x)$ ☐ D $-3 \cos(3x)$

Question 4**1 mark**

A pendulum of length 1.5 metres swings through an angle of 0.2 radians. Calculate the length of the arc traced by the pendulum.

- ☐ A 0.15 m
- ☐ B 0.45 m
- ☐ C 0.30 m
- ☐ D 7.5 m

Question 5**1 mark**

Which of the following describes the transformation required to obtain the graph of $y = \cos(x) - 2$ from the parent function $y = \cos(x)$?

- ☐ A Horizontal shift 2 units left
- ☐ B Horizontal shift 2 units right
- ☐ C Vertical shift 2 units up
- ☐ D Vertical shift 2 units down

Question 6**1 mark**

Express $\log_a x + \log_a y - 3 \log_a z$ as a single logarithm.

- ☐ A $\log_a(x + y - 3z)$
- ☐ B $\log_a\left(\frac{x+y}{z^3}\right)$
- ☐ C $\log_a\left(\frac{xy}{z^3}\right)$
- ☐ D $\log_a\left(\frac{xy}{3z}\right)$

Question 7

1 mark

What is the equivalent radian measure for an angle of 150° ?

☐ A $\frac{2\pi}{3}$

☐ B $\frac{5\pi}{3}$

☐ C $\frac{3\pi}{4}$

☐ D $\frac{5\pi}{6}$

Question 8**1 mark**

Evaluate the integral $\int \sin(x) dx$.

- ☐ A $-\sin(x) + C$
- ☐ B $-\cos(x) + C$
- ☐ C $\cos(x) + C$
- ☐ D $\sin(x) + C$

Question 9**1 mark**

What is the maximum value (amplitude) of the function $f(x) = 5 \cos(x) + 2$?

- ☐ A 7
- ☐ B 3
- ☐ C 2
- ☐ D 5

Question 10**1 mark**

Which equation represents the vertical asymptote of the function $y = \log_e(x - 4)$?

- ☐ A $x = 0$
- ☐ B $x = -4$
- ☐ C $y = 4$
- ☐ D $x = 4$

Section 2 / 3

20 total marks

Short Answer

Each question is worth 2 marks. Show all working to receive full credit.

Question 11

2 marks

Find the exact perimeter of a sector in a circle of radius 6 cm where the angle of the sector is $\frac{\pi}{3}$ radians.

Question 12

2 marks

State the domain and range of the function $f(x) = e^x + 4$.

Question 13

2 marks

Calculate the exact value of the area under the curve $y = \cos(x)$ between $x = 0$ and $x = \frac{\pi}{2}$.

Question 14**2 marks**

A function is defined as $f(x) = 2 \sin(x) + 3$. Sketch the graph of $f(x)$ for $0 \leq x \leq 2\pi$, clearly showing the y -intercept and the range of values.

Question 15**2 marks**

Solve for x in the equation $3e^{x-1} = 10$, giving your answer to 3 decimal places.

Question 16**2 marks**

Find the derivative of $y = x^2 \sin(x)$ with respect to x .

Question 17**2 marks**

Given a sector with radius r and arc length $s = 2r$, calculate the area of the sector in terms of r .

Question 18**2 marks**

A triangle OAB is formed within a circle of radius 6 cm, where O is the center and $\angle AOB = \frac{\pi}{3}$

radians. Calculate the area of the triangle OAB .

Question 19**2 marks**

Solve the equation $\log_{10}(x) + \log_{10}(x - 1) = 1$ for x . Show that only one solution is valid.

Question 20**2 marks**

Find the equation of the line that passes through $(2, 2)$ and is perpendicular to the line $y = 1$.

Long Answer

Each question is worth 5 marks. Comprehensive working and reasoning must be shown.

Question 21

5 total marks

An advertising logo is created using two intersecting circles. The first circle has its centre at the origin $O(0, 0)$ and a radius of 1 metre. The second circle also has a radius of 1 metre and its centre is located at $C(1, 0)$. The two circles intersect at points A and B . The region of interest is the total shaded area formed by the union of both circles.

A | Using the details provided about the advertising logo:

1. Show that the triangle OAC is an equilateral triangle.
2. Find the exact area of the sector OAB in the first circle.
3. Calculate the total area of the logo (the sum of the areas of both circles minus the area of their intersection).

Question 22

5 total marks

The height of the tide H (in metres) in a specific harbor over a 12-hour period is modeled by the function:

$$H(t) = 4 \cos\left(\frac{\pi}{6}t\right) + 6$$

where t is the time in hours since midnight ($0 \leq t \leq 12$).

A | Using the harbor tide model:

1. Determine the time at which the first low tide occurs.
2. A large ship can only enter the harbor if the water height is at least 8 metres. Find the time interval(s) between $t = 0$ and $t = 12$ during which the ship can safely enter the harbor.

Question 23

5 marks

A region is bounded by the curve

$$y = \sqrt{\sin(x)}$$

, the x -axis, and the vertical lines $x = 0$ and $x = \pi$.

1. Find the volume of the solid generated when this region is rotated 360° about the x -axis.
2. Find the rate of change of the area of the cross-section of this solid with respect to x at the point $x = \frac{\pi}{4}$.

Question 24

5 marks

The population P of a colony of bacteria after t hours is modeled by $P(t) = 100e^{kt}$.

1. If the population doubles in 5 hours, find the exact value of the growth constant k .
2. Determine the time t required for the population to reach 1000 bacteria. Give your answer to two decimal places.
3. Sketch the graph of $P(t)$ for $t \geq 0$, labeling the y -intercept and indicating the general shape of the growth.

End of Examination

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- View the complete answer key with detailed explanations



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- Detailed explanations for each answer